

BRENT A STATION 63113, United KingdomWMO: **996070**Lat: **61.000N**Lon: **1.708E**Elev: **30**StdP: **100.97**Time Zone: **0.00 (EUW)**Period: **02-15**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.1	1.0	N/A	N/A	N/A	N/A	N/A	N/A	22.0	3.5	20.5	4.1	9.6	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.2	18.9	N/A	17.4	N/A	16.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.4	15.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	5.5	5.2	5.6	6.9	8.6	11.1	13.3	14.1	12.3	9.5	7.9	5.9
	DBStd	3.64	1.71	2.03	2.03	1.92	2.11	2.08	1.98	1.83	1.72	1.91	1.77	2.25
	HDD10.0	788	139	133	137	93	54	9	0	0	2	31	63	127
	HDD18.3	3464	397	367	395	342	303	217	158	133	180	274	312	386
	CDD10.0	369	0	0	0	1	9	43	101	127	72	16	1	0
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.0	9.8	8.9	8.8	7.3	7.0	6.0	6.6	6.3	7.7	8.2	9.5	9.4
Precipitation	PrecAvg	1341	161	132	127	72	48	65	81	100	117	141	139	156
	PrecMax	1652	252	257	237	123	83	132	153	162	190	216	220	240
	PrecMin	954	76	58	46	26	22	26	40	66	48	58	43	65
	PrecStd	165	48	53	46	23	16	24	22	25	34	41	39	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	9.4	10.1	12.4	15.7	19.5	20.6	21.2	18.1	14.5	11.7	10.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	8.9	8.7	9.1	11.0	13.4	16.9	19.0	18.7	16.3	13.3	11.0	9.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	8.4	8.3	8.5	10.0	12.2	14.9	17.1	17.5	15.1	12.4	10.5	9.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	8.0	7.9	8.0	9.2	11.1	13.5	15.9	16.5	14.5	11.9	10.1	8.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.6	2.5	2.6	2.5	2.3	2.2	2.2	2.2	2.0	2.1	2.5	2.8
		MCDBR	2.9	2.8	2.7	3.6	3.9	4.2	4.2	3.8	2.6	2.2	2.9	3.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.334	0.355	0.368	0.380	0.370	0.362	0.380	0.370	0.364	0.346	0.321	0.388	
	taud	2.067	2.191	2.271	2.307	2.409	2.471	2.437	2.463	2.432	2.311	2.133	2.322	
	Ebn at Noon	350	599	741	810	854	868	839	818	749	610	367	224	
	Edh at Noon	45	72	91	106	103	99	100	90	78	63	41	29	
All-Sky Solar Radiation	RadAvg	0.20	0.69	1.83	3.56	4.96	5.06	4.59	3.65	2.22	1.04	0.31	0.12	
	RadStd	0.03	0.09	0.20	0.29	0.35	0.55	0.41	0.33	0.17	0.13	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	+0.65	+0.62	N/A	N/A	N/A	-102	-134	N/A	N/A

Nomenclature: See separate page